

Constrained Adaptive Per-Sample Clipping Predictor

WRN-16-1 / CIFAR-10 DP-SGD

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I. Theoretical optimization

Constants.

$$T = 510, \quad T' = 2T = 1020, \quad K = 8, \quad C_L = 0.25, \quad C_H = C_{\text{base}} = 1, \\ q = \frac{1}{3}, \quad (\varepsilon, \delta) = (8, 10^{-5}), \quad N = 45,000.$$

Causal policy. For update $t > K$ and sampled record i , let

$$x_{t,i} = \left(u_t, H_{t,i}, \frac{t}{T'} \right), \quad u_t = R_w \text{vec}(w_t) \in \mathbb{R}^{512},$$

$$R_w[a, b] \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1/512), \quad H_{t,i} = \{(z_i^\tau, \log(1 + \nu_i^\tau))\}_{\tau=t-K}^{t-1} \in \mathbb{R}^{K \times 4281}.$$

Here $z_i^\tau \in \mathbb{R}^{4275}$ is the fixed gradient projection and $\nu_i^\tau \in \mathbb{R}^6$ contains exact block norms. Define

$$p_{t,i} = \pi_\theta(x_{t,i}), \quad B_{t,i} \sim \text{Bernoulli}(p_{t,i}), \quad C_{t,i} = C_L B_{t,i} + C_H(1 - B_{t,i}).$$

For $t \leq K$, set $C_{t,i} = C_H$.

DP-SGD mechanism. With independent Poisson indicators $S_i^t \sim \text{Bernoulli}(q)$,

$$\bar{g}_t = \frac{1}{qN} \left[\sum_{i=1}^N S_i^t \text{clip}(g_i^t, C_{t,i}) + Z_t \right], \quad Z_t \sim \mathcal{N}(0, \sigma^2 C_H^2 I),$$

$$w_{t+1} = w_t - \eta_t \bar{g}_t, \quad \text{clip}(g, C) = g \min \left\{ 1, \frac{C}{\|g\|_2} \right\}.$$

σ is calibrated for $(q, T', \varepsilon, \delta)$; noise always uses $C_H = 1$.

Advantage-weighted policy loss. For predictor row j , with measured advantage $A_j = L_j^H - L_j^L$,

$$y_j = \mathbf{1}\{A_j > 0\}, \quad s_A = \text{median}_j |A_j| = 4.868945805 \times 10^{-5}, \quad \omega_j = \frac{|A_j|}{s_A + 10^{-12}},$$

$$\mathcal{L}_{\text{pred}}(\theta) = \frac{\sum_{j=1}^M \omega_j \text{BCE}(y_j, \pi_\theta(x_j))}{\sum_{j=1}^M \omega_j + 10^{-12}}.$$

Population clipping budget.

$$\frac{1}{N} \sum_{i=1}^N \mathbb{E}_{\pi, S} \left[\sum_{t=1}^{T'} S_i^t C_{t,i}^2 \right] \leq q T C_{\text{base}}^2 = q T.$$

The first K high-clipping updates consume K units. Hence the controlled horizon obeys

$$\frac{1}{T' - K} \sum_{t=K+1}^{T'} \mathbb{E}_\pi[C_{t,i}^2] \leq \beta_K, qquad \beta_K = \frac{T - K}{T' - K} = \frac{502}{1012} \approx 0.496047.$$

Since $\mathbb{E}[C^2] = C_L^2 p + C_H^2(1 - p)$, define

$$g(\theta) = \frac{1}{M} \sum_{j=1}^M [0.0625 p_j + (1 - p_j)] - \beta_K \leq 0, \quad p_j = \pi_\theta(x_j).$$

Equivalently, $M^{-1} \sum_j p_j \geq 0.53755$.

Constrained problem.

$$\boxed{\min_{\theta} \max_{\lambda \geq 0} \mathcal{L}_{\text{pred}}(\theta) + \lambda g(\theta)}$$

$$\theta \leftarrow \theta - \eta_\theta \nabla_{\theta} (\mathcal{L}_{\text{pred}} + \lambda g), \quad \lambda \leftarrow [\lambda + \eta_\lambda g(\theta)]_+.$$

II. Operational specification

Feature materialization.

1. Fix and record R_w , its seed, parameter ordering, and FP32 dtype. Project all 5,100 checkpoints to $u \in \mathbb{R}^{512}$.
2. For each labeled row with checkpoint $s \geq K$, retrieve the last K gradient sketches and block norms; group reads by (`run`, `step`).
3. Fit every feature transform on runs 0–7 only. Store sketches in BF16; convert minibatches to FP32. Exclude rows with $s < K$.

Predictor.

$$e_i^\tau = \text{GELU}(W_z z_i^\tau + W_\nu \log(1 + \nu_i^\tau) + b) \in \mathbb{R}^{128},$$

$$e_j^h = \text{GRU}_{128}(e_i^{s-K+1:s}), \quad e_j^w = \text{MLP}_{512 \rightarrow 128 \rightarrow 64}(u_s),$$

$$p_j = \text{sigmoid}(\text{MLP}_{193 \rightarrow 64 \rightarrow 1}[e_j^h, e_j^w, (s+1)/T']).$$

Split and training.

Split	Runs	Rows before warm-up removal	Rows with $s \geq 8$
Train	0–7	16,320	16,096
Validation	8	2,040	2,012
Test	9	2,040	2,012

For every epoch:

1. update θ by predictor minibatches;
2. evaluate $g(\theta)$ on the complete training split;
3. update λ once and project it to $[0, \lambda_{\max}]$;
4. retain feasible checkpoints, $g(\theta) \leq \epsilon_g$.

Select the feasible checkpoint maximizing validation policy value. Never tune on run 9.

Offline evaluation.

$$\widehat{V}(\pi) = \frac{1}{M} \sum_j A_j \mathbf{1}\{C_j = C_L\}, \quad \widehat{R}(\pi) = \frac{1}{M} \sum_j [\max(A_j, 0) - A_j \mathbf{1}\{C_j = C_L\}].$$

Evaluate (i) $C_j \sim \pi_\theta(\cdot | x_j)$ over repeated policy seeds and (ii) deterministic $C_j = C_L$ iff $p_j \geq \frac{1}{2}$. Report $\widehat{V}$, $\widehat{R}$, weighted accuracy, low-action rate, mean C^2 , and $g(\theta)$.

Online adaptive run. Maintain a rolling K -step history for all 45,000 private records. At each update: project w_t ; Poisson-sample; score sampled records; sample $C_{t,i}$; apply the DP-SGD update; sweep canonical private examples at w_{t+1} ; append projections and norms; discard history older than K . Probabilities, actions, gradients, and histories remain internal.

Paired end-to-end comparison.

	Basic DP-SGD	Predictor DP-SGD
Private train set	CIFAR-10 45,000	same
Public predictor set	unused for updates	CIFAR-10 5,000
Updates / sampling	$T' = 1020$, $q = 1/3$	same
Clipping	$C = 1$	$C = 1$ for $t \leq 8$; then π_θ
Noise	calibrated to $(8, 10^{-5})$	identical σ , $C_H = 1$
Randomness	paired seeds	paired seeds
Evaluation	CIFAR-10 test 10,000	same

Pair initialization, Poisson batches, augmentation, optimizer schedule, and Gaussian-noise streams. Primary endpoints: final/best EMA test accuracy and cross-entropy. Also report privacy accountants, learning curves, realized action rates, mean C^2 , budget violation, and runtime.